Remote Sensing KEY

2025 C Division

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Team Name/ Number: _____

Competitor Names: _____

Tiebreakers: Questions in Reverse Order

SCORE: ____ / 128

Section 1: Acronyms [20 pts, 1 each]

Expand each of the following acronyms completely. Each question is worth one point, inexact answers will receive no credit.

1. GPS: Global Positioning System
2. VTIR: Visible and Thermal Infrared Radiometer
3. ASTER: Advanced Spaceborne Thermal Emission and Reflection Radiometer
4. CZCS: Coastal Zone Color Scanner
5. ARVI: Atmospherically Resistant Vegetation Index
6. ADEOS: Advanced Earth Observing Satellite
7. CIESIN: Consortium for International Earth Science Information Network
8. HSRL: High Spectral Resolution Lidar
9. LVIS: Land, Vegetation, and Ice Sensor
10. MBES: Multibeam Echo Sounding
11. MIMR: Multifrequency Imaging Microwave Radiometer
12. TOLNet: Tropospheric Ozone Lidar Network
13. TSAVI: Transformed Soil Adjusted Vegetation Index
14. WDVI: Weighted Difference Vegetation Index
15. TBM: Terrestrial Biosphere Model
16. SSS: Sea Surface Salinity
17. RGB: Red-Green-Blue
18. POEMS: Positron Electron Magnet Spectrometer
19. S-GCOS: Space-Based Global Change Observation System

20. SaaS: Software as a Service

Section 2: Instrumentation & Physics [20 pts]

21. Explain the difference between passive and active sensors (2 pts):

    Active sensors emit energy to detect objects, while passive sensors detect energy naturally emitted or reflected by objects.

22. For each of the following, classify it as passive or active (6 pts, 1 each)

    a. Radar Active

    b. Sonar Active

    c. Cameras Passive

    d. Lidar Active

    e. Infrared Sensors Passive

    f. Radiometers Passive

23. Explain the difference between image quality and fidelity (2 pts)

    Image quality is the subject preference for one image over another, image fidelity represents the ability of a process to render an image accurately, without any visible distortion or information loss.

24. How is the peak wavelength of emitted radiation related to the absolute temperature of the black body? (2 pts)

    They are inversely proportional.

25. What wave property changes to cause refraction? (2 pts)

    Wave speed.

26. What is the gradual loss of force of a wave called? (2 pts)

    Attenuation.

27. What property does reflection off shiny surfaces have, and what is the technical name? (2 pts, one for property and one for name)

    The angle of incidence equals the angle of reflection, specular reflection.

28. What word describes the movement of a wave from one medium to another? (2 pts)

    Transmission.

Section 3: Interpretation [20 pts]

Refer to the image sheet for this question. For each of the five images, identify which satellite the image is taken from, and what the primary purpose of that satellite is. Each correct satellite name/ purpose will earn 2 points.

29. CERES on NPP, radiative energy flux
30. GRACE, sea level rise/ surface waves
31. Suomi NPP, optical, infrared, and Doppler radar imagery of clouds, precipitation, and extreme weather events.
32. OCO-2, distributions and radiative effects of greenhouse gases, trace atmospheric gases, and aerosols.

33. ATMS, atmospheric and sea surface temperatures.

Section 4: Climate Processes + Energy [20 pts, 2 each]

34. Which of the following is a primary characteristic of the El Niño phase of ENSO?

A) Unusually strong trade winds across the Pacific.

B) Warming of sea surface temperatures in the central and eastern equatorial Pacific.

C) Below-average rainfall in the southwestern United States.

D) Increased upwelling of cold, nutrient-rich waters off the coast of Peru.

35. During a La Niña event, what is typically observed in the eastern Pacific Ocean?

A) Weakening of the Walker Circulation.

B) Unusually cold sea surface temperatures.

C) An increase in hurricane activity in the eastern Pacific.

D) Reduced global precipitation.

36. The conversion of ammonia (NH_3) or ammonium (NH_4^+) into nitrites (NO_2^-) and then into nitrates (NO_3^-) by bacteria in the soil is known as:

A) Denitrification

B) Nitrogen fixation

C) Nitrification

D) Ammonification

37. Unlike the carbon and nitrogen cycles, the phosphorus cycle does not typically involve:

A) Biological absorption by plants.

B) Cycling through rocks and sediments.

C) Movement through aquatic ecosystems.

D) A significant atmospheric gaseous phase.

38. Which of the following represents the largest natural reservoir of carbon on Earth?

A) The atmosphere

B) Living biomass

C) The oceans

D) Sedimentary rocks and fossil fuels

39. What is the average albedo of the Earth?

0.3

40. Explain ice-albedo feedback.

As ice melts, the darker ocean is exposed, decreasing the albedo and the rate of ice melt.

41. What factors could lead to increased cloud albedo?

Increased water content, increased thickness, smaller droplet size.

42. Describe a climate feedback loop other than ice-albedo feedback.

Answers will vary greatly.

43. Explain the radiative forcing effects of aerosols.

Aerosols have mixed effects, with some such as sulfates causing cooling, while others such as soot cause warming.

Section 5: Mixed MCQ [20 pts, 2 each]

44. Which of the following statements are correct when describing the spectral signatures of vegetation and water?

A) Water and vegetation reflect somewhat similarly in the visible wavelengths

B) Similarity range is from about 0.4 to 0.7 nm but are almost always separable in the infrared

C) Both of the above

D) None of the above

45. ERS, Envisat, Sentinel and RISAT are example of which type of satellites:

A) Optical

B) Passive

C) Thermal

D) Microwave

46. At what wavelength will a planet with a surface temperature of 288 K radiate the most energy?

A) 45 nm

B) 11 μm

C) 53 nm

D) 90 μm

47. Which greenhouse gas is the primary cause of the breakdown of ozone?

A) Chlorofluorocarbon

B) Carbon dioxide

C) Methane

D) Nitrogen Trifluoride

48. Approximately what percent of the atmosphere is oxygen?

A) 78%

B) 21%

C) 26%

D) 1%

49. What color corresponds to the 380-450 nm range?

A) Red

B) Yellow

C) Green

D) Violet

50. What is the emissivity of a blackbody?

A) 1

B) 0.75

C) 0.25

D) 0

51. Cloud condensation nuclei are around what size?

A) 2 nanometers

B) 0.2 micrometers

C) 2 centimeters

D) 0.2 millimeters

52. Where are atmospheric CO2 levels measured?

A) French Alps

B) Key Largo

C) Mauna Loa

D) Maldives

53. Which of the following are false about atmospheric refraction?

A) Refraction in the atmosphere is caused by varying air densities at different heights

B) The refractive index of the atmosphere increases as altitude increases

C) Air pressure, temperature, and humidity are all factors that affect the atmosphere's refractivity

D) The refractive index of a layer in the atmosphere is the speed of light in a vacuum divided by the speed of light in that layer of the atmosphere

Section 5: Mixed FRQ [28 pts, 4 each]

Answers for this section will vary greatly, give credit for anything that makes sense.

54. Name and discuss at least two atmospheric or surface conditions that can lead to changes in equilibrium temperature of the Earth.

    Albedo, atmospheric gas concentrations.

55. Relate the concentrations of one greenhouse gas and one trace gas to natural influencing factors.

    Should discuss two gases, one in each category. Any non-human reasoning should be given credit.

56. Discuss an approach to combat climate change not related to reduce, reuse, recycle.

    Planting urban green spaces to combat heat islands.

57. Discuss 3 different aspects of image quality, and how each contributes to data collection.

Resolution - fine details, Contrast - object detection, Noise, accurate measurement.

58. Discuss techniques used to identify the composition of atmospheric gas.

Spectroscopy, gas chromatography, answer should elaborate.

59. An ideal blackbody radiator has a peak emission wavelength of 5.0×10^{-7} m. Determine the surface temperature of this blackbody.

Use Wien's Law, no partial credit, 5800K earns full points.

60. Consider the star Sirius A, which has a surface temperature of 9,667 degrees Celsius. Determine the total intensity of the star. Express your answer in $J/(s \cdot m^2)$.

Convert to Kelvin, 9940K, Use the Stefan-Boltzmann Law, $5.54 * 10^8$ earns full points.